

Liquid Bridges Mediate Crack Initiation, Propagation, and Coalescence to Suppress Film Failure within Flexible Electronics

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Recent flexible electronics with conformal interfaces between devices and human bodies are prone to receive circuit failure caused by uncontrollable cracking during physiological movements. A structural engineering strategy is reported that utilizes capillary-stabilized liquid bridges to spontaneously mediate crack initiation, propagation, and coalescence for film reinforcement. Specifically, rigid nanowire array are decorated onto flexible polydimethylsiloxane substrates and the nanoscale gaps between the nanowires are filled with non-volatile ionic liquids to form well-regulated meniscus. Using metal films as a model, it is found that stretchability of an Au film deposited on this meniscus exceeds that of its flat counterpart (180 vs 30%). In-situ optical observations and fluid dynamics analyses show that liquid bridge forces mechanically hinder the propagating of crack fronts and simultaneously initiate new cracks in different locations, leading to dispersed small cracks at strains below 80%. This scattering of cracks prevents the concentrated propagation and merging of cracks into penetrating fractures, with effective electrical percolation of Au films even under a high strain of 160%, which contrasts sharply with the counterparts without liquids where penetrating cracks occur at a small strain of $\approx 10\%$. Results indicate fluid mechanics as a versatile approach to reprogram film cracking for high-performance electronics.

1. Introduction

Flexible electronics, generally referred to as flex circuits or simply flexibles, are recently getting quite popular attentions because of their ability to create lightweight, bendable, and wearable electronic devices.^[1–8] Those devices are fabricated by coating electrically conductive films like copper,^[9,10] metal oxides,^[11,12] or amorphous silicon^[13] to integrate functional components on a flexible polymer substrate.^[5,7,14–22] For a typical example, metal film-based strain sensors, as an alternative for advanced capacitive strain sensors and so forth,^[23–25] represent a very specialized device to detect physical displacement, stretches, or compressions by conversion into changes in electrical resistance.^[26] While providing very sensitive monitoring, such sensors are faced with several challenges, including susceptibility to excessive strain.^[26–34] This process includes severe stretching/bending of the metal film, which may result in uncontrollable cracking. The cracks usually occur

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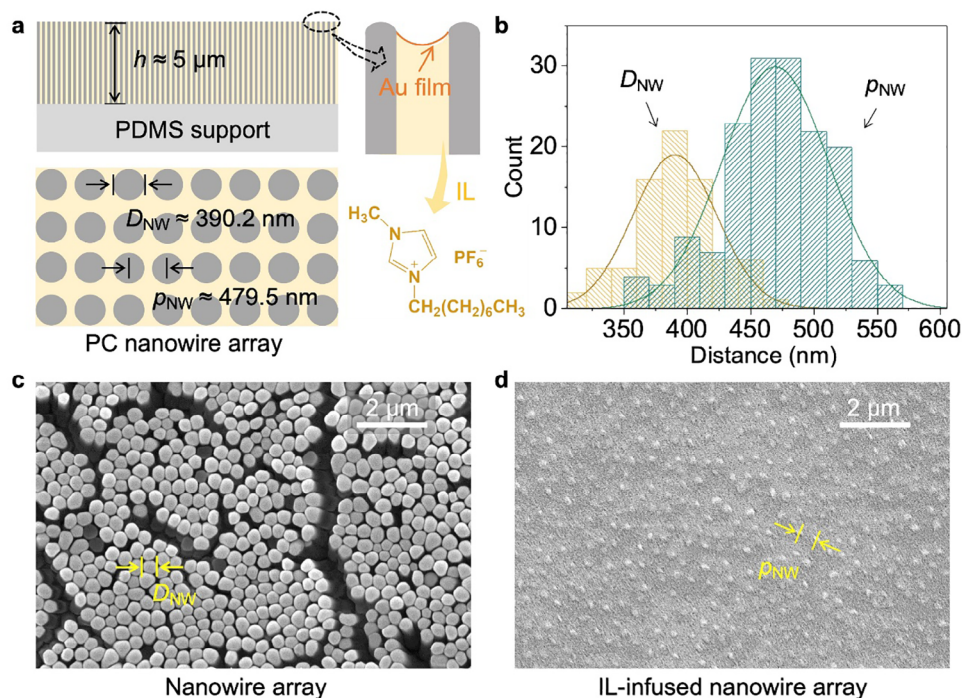


Figure 1. Design of a capillary-stabilized liquid film supporting an Au layer. a) Schematic illustration of the stereo structures, showing PC nanowires infused with IL and deposited with a thin Au layer on a flexible PDMS substrate. b) Distribution curves of the diameter of nanowires (D_{NW}) and center-to-center spacing (P_{NW}) in IL-filled PC nanowire arrays. c,d) SEM images of PC nanowire arrays before c) and after d) infusion of the IL.

along the direction perpendicular to stress and instantly extend across the entire metal film. These cracks show a low threshold of tolerable strains, defined as the maximum strain the sensor can endure without crack formation that would impair functionality, and may damage the main electrical circuit in flexible electronics.^[26,35–37] Importantly, it in turn constrains their inherent capability to monitor activities with large ranges over 30% uniaxial strain in general.

Liquid bridges (LBs) with a saddle shape, which capture small amounts of fluid between wettable surfaces, play a crucial role in drawing bridged solid surfaces together. LBs are essential in natural phenomena, for example, the wet grip shorebirds use LBs to catch insects and the retention of water between soil particles.^[38–48] Besides being important to nature, LBs also have various industrial applications, including ink distribution in printing and capillary action during soldering.^[49–54] Herein, we show the design in exploiting the LB forces between an elastomer-supported nanowire array to prevent crack growth and elicit coalescence, using Au films acting as a model functional film. Specifically, LB is formed at the intersection of air, nanowire wall, and ionic liquid (IL), known as the three-phase contact line (TCL), where Au films bonded to a capillary-stabilized IL begins to crack under tension. These intensive LB forces act to oppose and restrain continued crack propagation by distributing strain through the rigid nanowires and thus delaying the development of penetrating cracks, which is quite different with previous mechanism using geometric patterning (e.g., serpentine or fold structures) where their extended stretchability follows a buckling mechanism.^[4,6,9,55] In this work, using scanning electron microscopy (SEM) and optical microscopy, we observed

that at strains below $\approx 80\%$, numerous minor cracks appeared throughout the Au film. However, at higher strains (115–158%), the growth of these minor cracks between nanowires eventually coalesced, leading to the final formation of penetrating cracks. In sharp contrast, the results of the counterparts without nanowires or liquids showed many penetrating cracks at much lower strains than 10%. To demonstrate the practicality of this strategy, we integrated it into a strain sensor. This staged cracking mechanism enables both exceptional stretchability (up to 180% strain) and heightened sensitivity, with a Gauge Factor of ≈ 1031 (near $\approx 0\%$ strain) and $\approx 4 \times 10^8$ (near $\approx 180\%$ strain).

2. Results and Discussion

2.1. Capillary-Stabilized Fluidic Cushion that Supports Au Films

We used polycarbonate (PC) nanowire arrays as a model, which were $5 \mu\text{m}$ in height, filled with ionic liquid (IL, 1-octyl-3-methylimidazole hexafluorophosphate, C8) and supported on a flexible polydimethylsiloxane (PDMS) substrate (Figure 1a). Then, a $\approx 39 \text{ nm}$ thick Au film was deposited on the top surface of this structure (Figure S1, Supporting Information). The reason why we selected rigid PC nanowires (Figure S2, Supporting Information) is to enhance resistance of LB forces to the tensile forces among the nanowires during stretching. Our design concept is based on taking advantages of the LB forces among the nanowires to limit their separation and inhibit the propagation of cracks in the rigid film. The PC nanowire clusters were fabricated using a soft lithography molding method as follows.

First, a 25 wt.% PC solution was poured into an anodized aluminum oxide (AAO) template with a pore diameter of ≈ 390 nm, a center-to-center spacing of ≈ 450 nm, and a depth of 5 μm (Figure S3, Supporting Information). After solvent evaporation and the sample was solidified at room temperature for 20 h, the solid PC layer outside the pores was completely peeled off from the template, while the pores remained filled with PC. Next, a mixture of PDMS prepolymers and ethoxylated polyethylenimine (PEIE) was subsequently applied onto the surface of the template for a stretchable support after curing in an oven. Finally, bundles of closely-packed PC nanowires embedded in a flexible PDMS substrate were obtained after the removal of the AAO template (Figure S4, Supporting Information). PDMS was chosen as the substrate due to its excellent mechanical properties, chemical stability, and biocompatibility. In addition, the stretchability of this PDMS substrate was increased a lot by the incorporation of PEIE into it (Figure S5, Supporting Information). In the absence of IL, pure PC nanowires exhibited a bunch-like morphology with an average diameter of nanowires of ≈ 390 nm and a depth of about 5 μm (Figure 1c; Figures S6 and S7, Supporting Information). Size statistics indicate that the IL-infused nanowires have an average diameter (D_{NW}) of ≈ 390 nm, a center-to-center spacing (p_{NW}) between the nanowire tips of ≈ 479 nm (Figure 1a,b), which closely match the dimensions of the AAO template.

The PC nanowire arrays showed contact angles (CA) of $\approx 0^\circ$ towards IL-ethanol solutions with IL concentrations of 0–400 mg mL^{-1} and CA of $\approx 24^\circ$ towards pure IL (Figure S8, Supporting Information), demonstrating superhydrophilicity towards IL-ethanol solutions. When the IL-ethanol solutions were deposited on the PC nanowires, IL permeated into the space between the nanowire arrays with a meniscus during evaporation of the ethanol solvent (Figure S9, Supporting Information). The SEM images vividly show that the IL film is confined between the nanowires, which effectively supports an Au layer (Figure 1d; Figures S10 and S11, Supporting Information). Au films and Au elements are demonstrated to homogeneously distribute on the surface of IL-infused nanowire arrays (Figures S12–S14, Supporting Information). The ultralow vapor pressure of IL allows observing the IL film in a low-pressure SEM chamber. These results confirm that the liquids to support an Au layer were successfully prepared by capillary stabilization.

2.2. Fluidic Cushion Regulated Crack Initiation, Propagation, and Coalescence

To further validate the hypothesis that capillary-stabilized IL can regulate cracking of the metal films it support, we conducted further experimental measurements under various tension. These measurements describe the mechanism of this phased cracking process and provide greater insights into the strain versus electrical resistance relationship for Au film-coated liquid surfaces. We captured the morphological changes of the control group Au film on PDMS-supported PC nanowire arrays under continuous cyclic stretch in a strain range from 0 to 26%. As shown in Figure 2a, even Au films without inherent cracks before extension residence on the surface of continuous IL which is attributed to the dominant effects of surface tension between neighboring nanowires with nanoscale gaps. Initial cracks ap-

pear at random locations on the Au film subject to a small strain of $\approx 10\%$. With further increase in strain, these cracks rapidly spread in a direction perpendicular to the applied stretch, eventually covering the entire film, as also evidenced by a substantial rise in electrical resistance (refer to details below). Similar perpendicular penetrating cracks also occur in Au films on a flat PDMS substrate when apply a small strain of $\approx 19\%$ (Figure S15, Supporting Information). Figure 2b is corresponding structural model of Figure 2a, which vividly exhibits that no LB form among nanowires.

Next, uniaxial tensile strain was applied to the substrate for Au films on nanowire-supported IL layers at a concentration of 120 mg mL^{-1} (Figure 2c). In-situ optical microscopy was utilized to monitor the cracking progression. Minor cracks often took the lead during uniaxial tensile strain along the substrate, whose crack length lay between 1 and 30 μm at 80% strain (Figure 2c,e). This is in sharp contrast to the counterparts without infused liquids where penetrating cracks developed at a strain level as low as 10%. When the strain level reaches a critical value of $\approx 115\%$, a secondary crack propagation stage starts from the pretrained cracks without the development of new ones. These large cracks grew further and was eventually coalesced to develop the penetrating cracks in the direction perpendicular to the applied strain at $\approx 160\%$ strain (Figure 2c,e). For example, as illustrated in Figure 2c, cracks LC1 and LC2 expand and merge to form crack LC12 prior to reaching 160% strain. All these observations suggest that the IL layer behaves like a fluidic buffer to effectively control the cracking process of the Au film and postpone the generation of penetrating cracks. The structural model of the IL infused-nanowires further clearly delineates that the LB forms among nanowires (Figure 2d). Figure 2f summarize the relationship between C_L and ϵ of 120 mg mL^{-1} IL-infused nanowires and bare nanowires (the inset). With the increased strains, the C_L of both samples increases evidently and distributes more widely. In the case of pure nanowire structures subjected to tensile strains of 10, 20, and 30%, pervasive large cracks exceeding 200 μm in length are observed on the surface. Conversely, for the 120 mg mL^{-1} IL-infused nanowires, the maximum C_L remains below 190 μm even under the maximum strain of 160%.

Figures S16 and S17 (Supporting Information) summarize statistical analyses of the distribution of the crack size including C_L and crack width (C_W) of the 120 mg mL^{-1} IL-infused nanowires under different stretching degrees. No cracks are observed for the as-received sample (Figure 2c,e), however, when the sample is strained up to 80%, several short cracks appear with lengths less than 40 μm . At strain of 115%, short cracks tend to expand and merge, and large penetrating cracks ≈ 80 –120 μm form in this case. With the increase in strain up to 160%, growth and interlinking of cracks become more intense. More large cracked regions appear, particularly in the range of ≈ 120 –180 μm with an increase in population. With the increased strain from 0 to about 160%, the C_W increases from 0 to ≈ 4 –6 μm .

As mentioned above, our goal is to control large-size crack formation to develop a highly stretchable and sensitive device based on metal film. To this end, we explore the relationship between the electrical properties of the Au film and the structural characteristics of the cracks. The results of our electrical conductivity

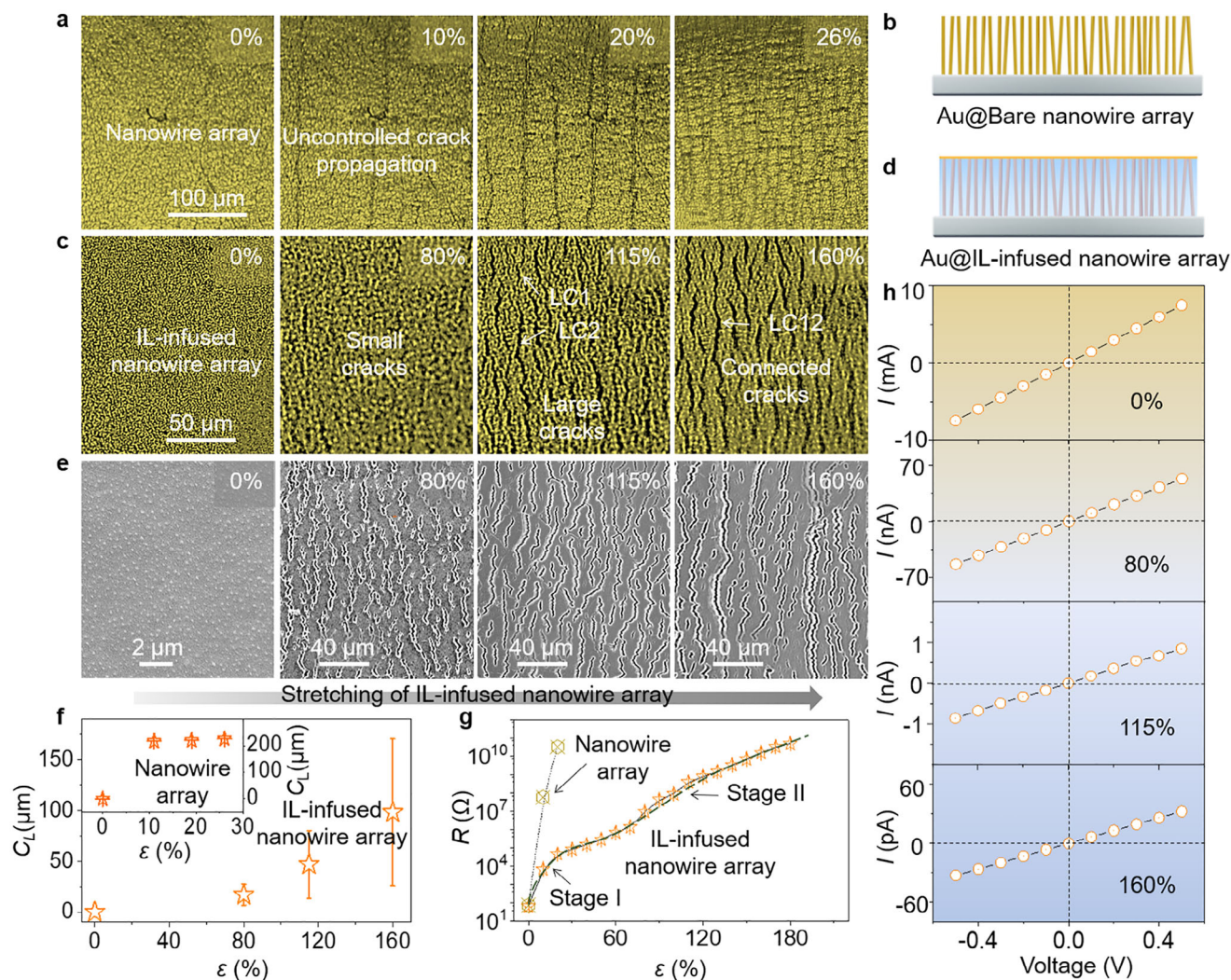


Figure 2. Effects of strain on topography variations of Au film deposited IL-infused nanowires-based sensors and electrical resistance. a–d) In-situ optical images capturing the cracking processes of Au films on bare nanowire structures without IL (a) and IL-infused (120 mg mL^{-1}) nanowires (c), and their corresponding structural models (b,d). e) Corresponding SEM images displaying the cracking behaviors of Au films on IL-infused nanowires (120 mg mL^{-1}) under strains of 0, 80, 115, and 160%, respectively. f) The relationship between crack length (C_L) and strain (ϵ) of IL-infused nanowire array and bare nanowire array (the inset). g) Electrical resistance of Au films on bare nanowires and IL-infused nanowires as a function of strain. h) Voltage versus current plots under strains of 0, 80, 115, and 160%, respectively.

measurements are presented in Figure 2g,h. Our studies show that, within the applicable strain range from 0 to 160%, the Au film can keep a stable ohmic behavior, supported by liquid, which demonstrates that the liquid can minimally affect the overall conductivity of the conductive layer. Notably, the resistance exhibits a more rapid increase within the strain ranges of 0–60% and 90–120%, which aligned with our in-situ observations of the phased process of crack initiation and propagation. The presented results contrast sharply with those obtained from Au films that are deposited directly onto bare nanowire arrays, which fail under strains of $\approx 30\%$. Such a contrast reflects the huge contribution of liquid support in enhancing flexible electronics performance. In that case, the liquid would stabilize the Au film and enhance its stretchability. This is why functionality in the device would stay well at higher strains.

2.3. LB Mediate Cracking of Au Film on a Fluidic Cushion

To illustrate how the fluidic cushion regulates crack growth, we trace the crack tips using SEM (Figure 3a). We can clearly see that the fluids confined among the nanowires develop a meniscus, which bridges the neighboring nanowires. It means there are extra forces applied on NW-1 and NW-2 compared to nanowires without meniscus—what we call LB force.

We performed an in-depth analysis of the shapes of the nanowires and the fluids they contained (Figure 3b) to define the LB force (F_{LB}). Disregarding the effects of gravity and surface contact at the top and bottom, we assume that the radius of the meniscus surface remains uniform along its length. The apparent CA ($\angle PDQ$ or θ_1) between the IL and the nanowire is observed, which is different with that of flat sample (Figure S19,

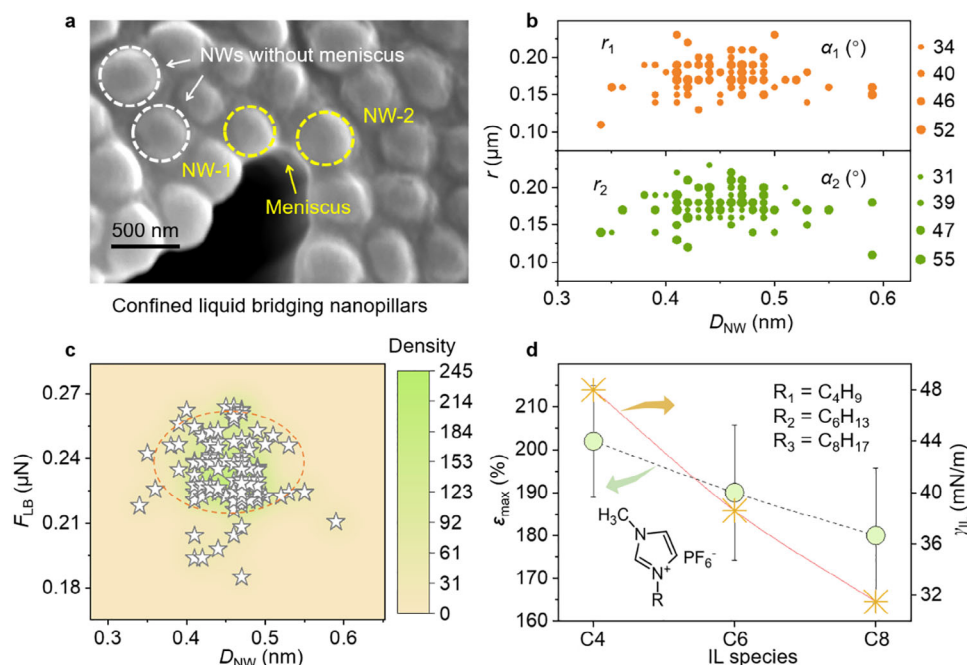


Figure 3. Liquid bridging adjacent nanowires forms a meniscus and provides additional LB attraction to suppress crack growth. a) SEM image capturing the cracking of Au films on nanowires infused with IL. b) Statistical parameters of the cracks (Figure S18, Supporting Information). c) Calculated LB attraction acting on adjacent nanowires at the crack tips. d) Effect of liquid's surface tension on the maximum strain of our electronic devices.

Supporting Information). The radius of the nanowire is represented by OD (r_1 and r_2), the radius of the meniscus liquid surface by O_3D (R_m), and the center-to-center distance between the two nanowires by O_1O_2 (d_{NW}).

$$\alpha_1 = \pi/2 - \theta_1 - \varphi_1 \quad (1)$$

F_{LB} is divided to two parts: the surface tension force (F_{ST}) and the Young-Laplace pressure force (F_L). Because the LB is not an independent object, the Young-Laplace pressure is negligible and the force acts on the entirety of the bundle of nanowires.^[56,57] Hence:

$$F_{LB} = F_{ST} = \gamma_{IL} h_{nw} \cos \alpha \quad (2)$$

Figure 3c shows that the F_{LB} is ≈ 0.24 μN. Assuming this force acts on the cross-sectional area of the nanowires, the LB-induced attraction (F_{LB}/A_{NW}) is 0.12 MPa after calculation. The presence of this additional force makes the propagation across the meniscus even less favorable compared to the creation of new cracks outside of those areas where there is a meniscus.

In addition to the C8 we selected for the concern of its low toxicity and chemical stability for applications in wearable devices, we also tested other ILs, including 1-butyl-3-methylimidazolium hexafluorophosphate (C4) and 1-hexyl-3-methylimidazolium hexafluorophosphate (C6), as the fluidic cushion. According to the data extracted from previous research, it can be seen from Figure 3d that with the increase of carbon chain length, the surface tension of bulk IL decreases. Meanwhile, the maximum strain exhibits a similar trend, which shows that the maximum strain increases with improved F_{LB} . The above

results highly confirm that the observed LB force may play an important role in regulating cracking of the supported Au film.

2.4. Impact of Reduced Liquids within Fluidic Cushions on the Evolution of Electrical Resistance in Au Films

To investigate how the IL fluidic cushion influences the changes in electrical resistance due to strain, we chosen two control samples by changing the concentration of IL infused into the nanowires. For the first control sample (cushion-A), excess IL was introduced inside the IL infusion using a concentration of 400 mg mL⁻¹. For another sample, we decreased the IL/ethanol concentration to 50 mg mL⁻¹, which was marked as cushion-B. We achieved this variation by modulating the concentrations of IL in ethanol during the sample preparation process (see Experimental Section).

We observed the topographic changes in the control group Au film on cushion-A within a strain range of 0 to 25%. As shown in Figure 4a, initial cracks formed randomly on the Au film when subjected to a minor strain of $\approx 15\%$. As the strain increased further, these cracks quickly propagated perpendicularly to the direction of the applied tension, eventually spreading throughout the entire film. Therefore, the electrical resistance was high. In this case, the fluidic cushion failure is due to the excess amount of IL, which submerge the nanowires completely, preventing the formation of LBs as the TCLs are lacked (Figures S20 and S21, Supporting Information). The corresponding structural models (Figure 4b) also vividly present the details of cracks, which accords very well with the reality. However, when the IL content is low (Figure 4c,d), large crack clusters appear on the

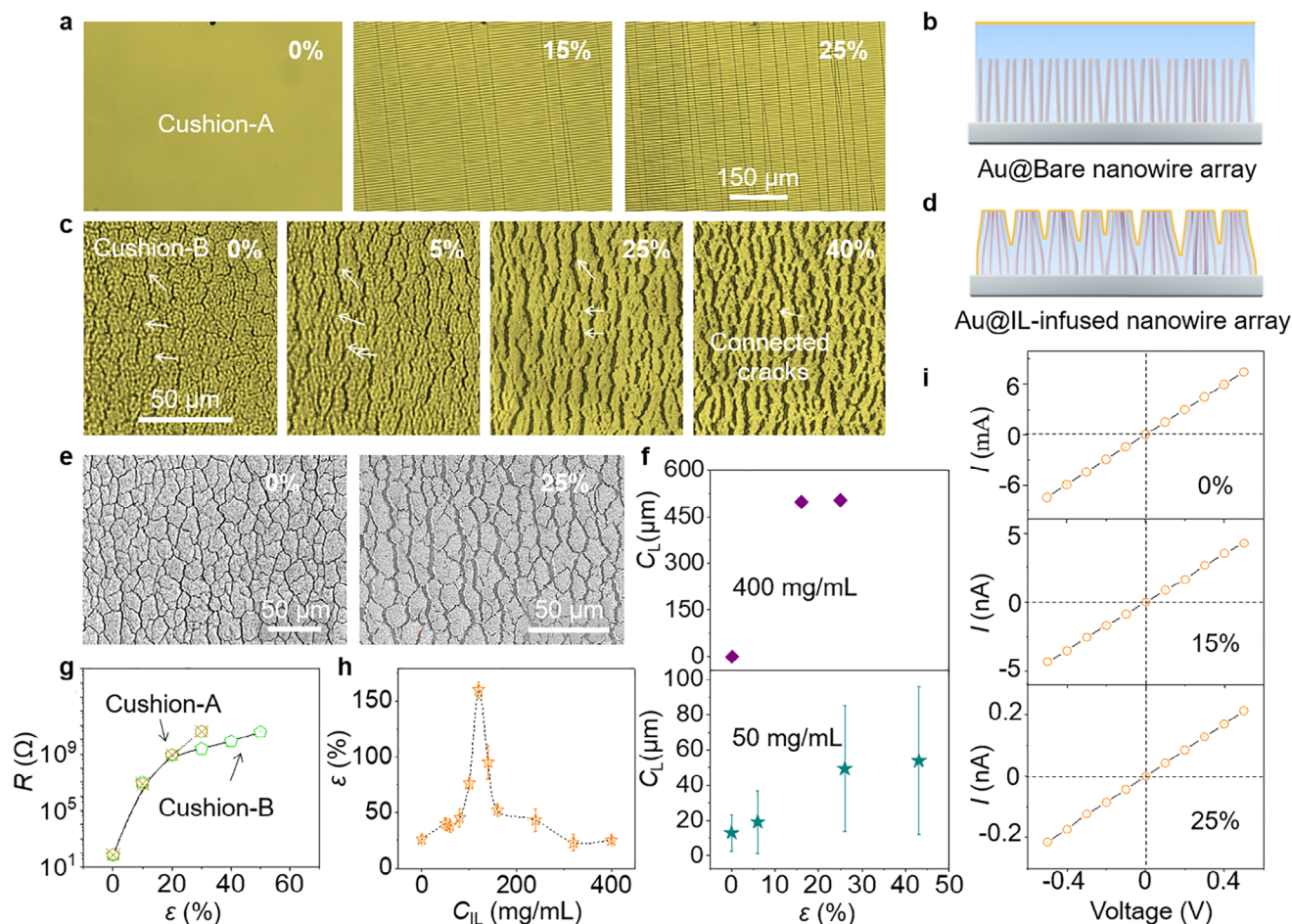


Figure 4. Influence of IL concentration (C_{IL}) on the surface topography and electrical resistance of Au films on IL-infused nanowires-based sensors. a–d) In-situ optical imaging of the cracking process of Au film deposited on (a) a nanowire array with excessive C_{IL} of 400 mg mL⁻¹ (cushion-A, strain range of 0–25%) and (c) a nanowire array with much less C_{IL} of 50 mg mL⁻¹ (cushion-B, strain range of 0–40%), and their corresponding structural models (b, d). e) Corresponding SEM images of the nanowire bundles with lower C_{IL} content of 50 mg mL⁻¹ under various strains (0 and 25%). f) The variations of C_L of 400 and 50 mg mL⁻¹ IL-infused nanowires with the strains. g) Graph illustrating the variation in resistance of the strain sensor with strain for both cushion-A and cushion-B. h) Variations of the maximum strain ε for IL-infused nanowires with various C_{IL} ranged from 0–400 mg mL⁻¹. i) Voltage versus current measurements of Au film on cushion-B at 0, 15, and 25% strains, respectively.

unstressed surface instead of forming a dense and uniform structure (Figures S22 and S23, Supporting Information). This is due to the capillary forces during ethanol evaporation, which mechanically bundle the nanowires. Upon applying strain (0 to 25%), the pre-existing cracks tend to propagate and merge, leading to the rapid formation of largely penetrating cracks at a 25% strain, as illustrated in SEM images shown in Figure 4e.

Figure 4f summarized the C_L distribution of 400 and 50 mg mL⁻¹ IL-infused nanowires during the stretching process. When the strain increases, the C_L starts to increase, and their distribution becomes widespread. The 400 mg mL⁻¹ sample shows pervasively penetrating cracks with length $\approx$ 400–500 μ m when strain was $\approx$ 15%, while the amounts of penetrating cracks are increased with the increased strains (Figure 4a).

In the case of 50 mg mL⁻¹ IL-infused nanowires, most of the C_L in the unstressed sample ($\varepsilon = 0\%$) are smaller than 20 μ m. At a strain of 5%, larger cracks in the length range of 20 to 50 μ m

begin to appear. With a 25% strain, the crack growth and interconnection has become much more pronounced and the number of large cracks, mostly in the length range from 40 to 90 μ m, has increased. At a 40% strain, the surface is largely covered by penetrating large cracks between 40–110 μ m length. C_w also change from 1–2 to 4–6 μ m with the strains from 0 to 40% (Figures S24 and S25, Supporting Information). These findings suggest that the design in the shape of the fluidic cushion is important in dictating the cracking process. As indicated in Figure 4h, the tolerable strains firstly show upward trend with increased C_{IL} until reach the peak at 120 mg mL⁻¹, and then exhibit a downward trend as the C_{IL} further rises, suggesting that the sample achieve best performance at C_{IL} of 120 mg mL⁻¹. The SEM images under other C_{IL} indicate that the nanowire clusters become concentrated with increased C_{IL} until that uniform IL layer form among nanowires at C_{IL} of 120 mg mL⁻¹ (Figure S26, Supporting Information). However, when C_{IL} is larger than 140 mg mL⁻¹, only IL film layer rather than nanowire tip is observed. These results

prove that no meniscus LB force form among nanowires when C_{IL} is either too low or too high.

To evaluate the performance of the Au film under strain, we tested the electrical properties on both cushion-A and cushion-B. Results of our conductivity measurements are presented in Figure 4g,i. Our analyses indicate that the Au film on cushion-B shows stable ohmic behavior irrespective of applied strains, which means the shape phenomenon of the liquid minimally influences its ohmic characteristics. However, the resistance increases much more rapidly within the range of 0–40% strain for cushion-A and 0–60% strain for cushion-B, consistent with the in-situ observations of these cracking processes as shown in Figure 4a,c. These findings are in stark contrast to those of Au films deposited on well-designed fluidic cushions shown in Figure 2a, which fail at strains of $\approx 10\%$. This disparity highlights the key role of liquid support in enhancing the performance of flexible electronics.

2.5. Applications as Wearable Electronics

We conclude this study by highlighting the soft electronic devices that incorporated metal and IL film. For comparison, traditional carbon-based or ionic strain sensors with resistance change governed by geometric variation during deformation only show gauge factors of about 2.^[58,59] Markedly enhanced sensitivity of this gold film and its favorable stretchability make the devices highly suitable for applications in soft robotic actuation and the monitoring of the human body. The Au thin film deposited on the IL-infused bundles of nanowires shows excellent sensitivity against deformation in a broad range of strains. In detail, as shown in Figure 5a,b, the sensor reliably detects strains as small as 0.005%, under both minimal (0%) and maximal (100%) strain conditions. The six steps of strain magnitude correspond to each other, with each step being 0.005%. The corresponding gauge factors ($\Delta R/(\epsilon R_0)$) are 1045 and 1.3×10^6 , where ϵ denotes the applied strain. These results confirm that high sensitivity over a very large strain range. This Au thin film-coated, IL-infused nanowire-structured sensor is readily achieved performance superior to those in reported sensors using metals or other conductive materials presently (Figure S27, Supporting Information).

The sensor not only shows extraordinary sensitivity and stretchability, but also exhibits remarkable stability in a range of environmental conditions regarding temperature and humidity. It maintains consistent resistance in the temperature range from 25 to 45 °C (Figure 5c), with humidity having minimal influence on its performance. As illustrated in Figure 5d, for a relatively wide range of humidity levels ranging between 15 to 95%, the designed electronics show highly stable electric resistance. These findings suggest that this liquid-based sensor has great potential for real-world applications since its performance is reliable in various environmental aspects.

Based on the remarkable sensitivity, stretchability, and stability under a range of environmental conditions, we propose the strategy as a versatile and efficient way of fabricating highly sensitive and ultra-stretchable soft electronics. It holds broad potential applications in the detection of large and small deformations. In this work, we successfully integrated the sensors into soft robotic systems for monitoring various types of deformations. We first

mounted the sensors onto the surface of a pneumatic micro-flexible gripper designed to grasp thin sheets with 500 μm thickness (Figure 5e). The sensor exhibits high sensitivity throughout the grasping process, with its resistance rising from 0.07 to 0.23 k Ω . Upon completion of the grasp, resistance settled to ≈ 0.17 k Ω . The resistance returns from 0.20 to 0.08 k Ω after release when tested during the release process, shown in Figure 5f, proving the sensor can detect fine deformation in practical scenarios. Moreover, the sensors are attached to the surface of a large-scale electric flexible robotic arm gripper to grasp larger objects with dimensions of 5 cm in length and width. As shown in Figure 5g, the resistance increased from 0.1 to 12.89 k Ω after grasping, and finally stabilized at 8.36 k Ω when the grasp was completed. Upon release, the resistance recovered from 9.88 k Ω to the initial value of 0.34 k Ω (Figure 5h). All these results prove that our sensors can detect both macro as well as micro-scale deformation modes, making them suitable for practical applications.

Apart from the robotic applications, our sensors can be effectively applied to human skin for precise detection of various motion types: finger vertical bending (Figure 5i), finger horizontal bending (Figure 5j), and wrist bending (Figure 5k). For example, as shown in Figure 5i, when vertical bending of a finger reaches 10°, the sensor stably and sensitively detects the degree of bending, with resistance increasing linearly from 0.11 to 0.74 k Ω . Regarding other basic motions such as lateral finger bending and wrist flexion, the performance of strain gauges on the sensors demonstrates excellent sensitivity and stability (Figure 5j,k). Additionally, the sensors can detect the amplitude of various motions accurately. When the bending angle increases from 4 to 6° and then to 10°, the resistance increases from 0.41 to 0.54 and finally to 0.74 k Ω . These responses are fully reversible, with resistances decreasing from 0.71 to 0.54 and further to 0.39 k Ω , allowing for repeated and reliable detections (Figure 5l). The sensors also exhibit very good sensitivity and stability while monitoring large-scale vertical bending from 40 to 90° (Figure 5m). Further, the sensors show the transition of various types of movements (Figure 5n). For example, a 15° vertical bend at 1.25 k Ω can be converted into a 20° horizontal bend where the resistance changes to 2.50 k Ω . This behavior results in very good repeatability and stability. These results point to the fact that the sensors show supreme sensitivity and stability in the detection of complicated human motions, and therefore are of immense potential in applications concerning human motion monitoring and soft robotic monitoring.

3. Conclusion

In this study, we propose a capillary-stabilized liquid bridge (LB) strategy within nanowire arrays to control crack initiation, propagation, and coalescence in flexible electronics. By filling nanoscale gaps between rigid polycarbonate (PC) nanowires with non-volatile ionic liquids (ILs), we achieved well-controlled menisci that support the Au film, forming LB forces, which greatly impeded crack propagation while promoting dispersed microcracks, preventing catastrophic film failure even under high strain. This approach effectively suppresses film fracture and offers a promising pathway for developing high-performance flexible electronics with exceptional mechanical robustness and functional reliability for complex motion detection, enabling both

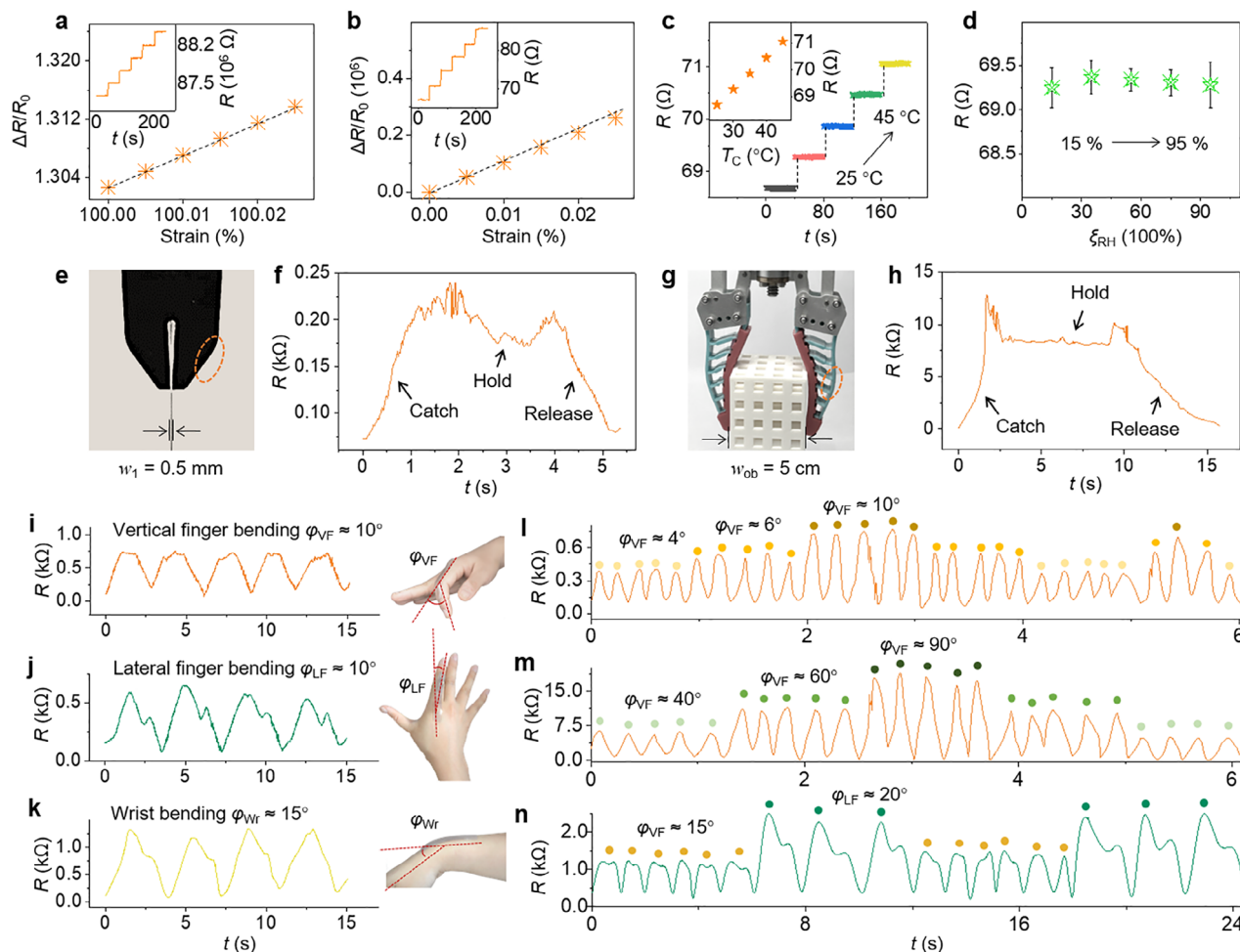


Figure 5. Design and characterization of ultrastretchable and highly sensitive electronic devices. a,b) $\Delta R/R_0$ of Au films as a function of strain over a strain range of a) 100–100.025%, or b) 0–0.025% with each step forward of 0.005% strain (insets). The insets are their corresponding step-strain experiments. c) Stability testing of sensor resistance at diverse operating temperatures. The inset is corresponding resistance vs working temperature. d) Performance of the designed sensors under various environmental humidity test. e) The actuator integrated with our designed sensors holds a 0.5 mm thick sheet. f) The electronic resistance in relation to actuator deformation during the catch and release processes of the sheet. g) A large soft mechanical gripper equipped with the designed sensors is shown holding a cube (5 cm in length and width). h) Variations in electrical resistance during the process of catching, holding, and releasing the cube by the soft gripper. i–k) The resistance of the designed sensor changes in response to finger and wrist movements at various angles and in different directions. l–n) The sensor is capable of detecting both large and small finger bending movements, as well as transitions between different bending behaviors.

exceptional stretchability (up to 180% strain) and heightened sensitivity with a Gauge Factor of ≈ 1031 ($\approx 0\%$ strain) and $\approx 4 \times 10^8$ ($\approx 180\%$ strain). Future work may consider how relevant this design strategy is to a broader range of materials and device architectures, such as other types of conductive/semiconductive films and flexible substrates. Moreover, long-term stability and durability of the proposed liquid-infused nanostructures under cyclic loading and other environmental conditions that are crucial for real application should be carefully investigated.

4. Experimental Section

Materials: Polycarbonate (PC) with a molecular weight range of 20 000–70 000 g mol^{-1} (Baichuang Shujiao Materials Ltd.) was utilized for fab-

ricating the nanowire arrays. Anodized aluminum oxide (AAO) templates with pore diameters of ≈ 390 – 400 nm, center-to-center spacing of about 450 nm, and a depth of 5 μm were purchased from Shanghai ShangMu Nano Technology, Ltd. Polydimethylsiloxane (PDMS) prepolymer and its curing agent were obtained from Dow Corning. Ethoxylated polyethyleneimine (PEIE, 80% v/v; molecular weight = 70 000 g mol^{-1} ; Sigma-Aldrich) was incorporated to enhance substrate elasticity. Ionic liquids (ILs) including 1-octyl-3-methylimidazolium hexafluorophosphate (C8), 1-hexyl-3-methylimidazolium hexafluorophosphate (C6), and 1-butyl-3-methylimidazolium hexafluorophosphate (C4) were sourced from Sigma-Aldrich.

Preparation of Rigid Nanowire Arrays on Flexible Supports: The clusters of PC nanowires were fabricated using a soft lithography molding technique with AAO templates. A 25 wt.% solution of PC was prepared in dichloromethane by dissolving the polymer under continuous stirring until complete dissolution was ensured. This viscous solution was then carefully poured into the AAO templates, ensuring complete infiltration of the

pores. From SEM analyses, the templates had pore diameters of ≈ 390 nm, center-to-center spacing of 450 nm, and a depth of around 5 μm , as depicted in Figure S3 (Supporting Information).

These filled templates were kept at room temperature for 20 h to allow the evaporation of solvent and subsequent solidification of PC within the pores. After solidification, the excess layer of PC formed atop the template was carefully peeled off, preserving the integrity of the PC-filled pores.

Accordingly, a PDMS mixture was prepared by mixing 10 g of PDMS prepolymer with 1 g of curing agent, keeping the weight ratio at 10:1. To increase the mechanical stretchability of the PDMS substrate, 20 μL of PEIE (80% v/v in water; molecular weight = 70 000 g mol^{-1}) was added to the mixture. The resulting mixture was stirred for 20 min and then degassed under vacuum to remove any entrapped air bubbles. The PDMS-PEIE mixture was uniformly spread over the surface of the AAO template (0.2 mL in total). The assembly was cured in an oven at 90 $^{\circ}\text{C}$ for 3 h to allow cross-linking of the PDMS.

The extraction of the PC nanowires from the PDMS matrix was carried out by immersing the setup in a 2 M HCl solution maintained at 80 $^{\circ}\text{C}$ for 5 h. During the acid treatment, the AAO template was dissolved without affecting the PC nanowires or the PDMS substrate. The samples were then treated several times with deionized water and ethanol to remove residual acid and contaminants after the template removal. The final product consisted of bundles of densely packed PC nanowires anchored within the flexible PDMS substrate. The structure and dimension of the nanowires were confirmed by SEM imaging, showing an average diameter of ≈ 390 nm and a length of ≈ 5 μm , as depicted in Figure 1c (Figures S6 and S7, Supporting Information). Mechanical test results showed that the addition of PEIE remarkably increased the elasticity of the PDMS substrate (Figure S5, Supporting Information).

Fabrication of IL Film-Based Strain Sensors: IL film-based strain sensors were fabricated by immersing PC-PDMS hybrid samples into ethanol solutions with different IL concentrations. Specifically, the samples were immersed in solutions of 1-octyl-3-methylimidazolium hexafluorophosphate (C8) at concentrations ranging from 0 to 400 mg mL^{-1} (the mass of IL per 1 mL of ethanol). The immersion duration was maintained at 5 min to facilitate IL infiltration into the interstitial spaces between the PC nanowires. The samples were then taken out from IL solutions and air-dried at room temperature for 1 h to evaporate the ethanol solvent. Consequently, capillary-stabilized liquid films were formed between nearby nanowires due to the confinement of IL inside the nanostructures and the strong capillary forces acting at the nanoscale.

After that, a thin layer of Au with thickness ≈ 39 nm was deposited onto the surface of IL-infused nanowire arrays via physical vapor deposition methods like sputtering or thermal evaporation. The Au coating represented the conductive layer of the strain sensors. Further studies of IL property variations on sensor performance were carried out using different ILs with the same fabrication protocol: 1-hexyl-3-methylimidazolium hexafluorophosphate (C6) and 1-butyl-3-methylimidazolium hexafluorophosphate (C4).

Characterization: The morphological and structural characteristics of the samples were examined using a FEI Nova NanoSEM 450 scanning electron microscope operating at an acceleration voltage of 10 kV. High-resolution SEM images were obtained to observe the nanowire arrays at various stages: post-fabrication, post-IL infusion, and post-Au deposition. Elemental mapping and energy-dispersive X-ray spectroscopy (EDX) analyses were conducted to confirm the uniform distribution of Au and to verify the elemental composition of the samples (Figures S12–S14, Supporting Information).

Contact angle (CA) measurements were performed using an OCA35 goniometer (Data Physics, Germany) to assess the wettability of the nanowire arrays with respect to different liquids. The apparent CA values were averaged over five measurements taken at distinct locations on each sample to ensure statistical reliability.

The Au film strain sensors were stretched while their response was monitored in real time by an Olympus BX51 optical microscope fitted with a digital camera and interfaced with a computer. This setup allowed for in situ observation of crack initiation, propagation, and coalescence in the Au

film under applied mechanical strain, providing insights into the mechanisms governing sensor performance.

X-ray photoelectron spectroscopy (XPS) was used to investigate the chemical composition of the samples. All measurements in this investigation were made using a Shimadzu PHI 250XI, allowing the detection and quantification of Au, PC, and IL components.

The mechanical tests of the PDMS substrates and the assembled strain sensors were conducted on an electronic universal testing machine, CMT6103, ZWICK, Instron 5969. Tests were performed at room temperature with a crosshead speed of 1 mm min^{-1} . The tensile samples were prepared in conformity with ISO 527-1:2012, using molds with a gauge length of 75 mm, a central width of 10 mm, a thickness of 4 mm, and an overall length of 170 mm. These molds were filled with PDMS mixtures and cured at 90 $^{\circ}\text{C}$ for 3 h. After curing, the specimens were demolded and subjected to tensile tests, with the deformation process recorded using a Nikon D7200 digital camera.

Measurement of Electrical Signals: The electrical properties of the strain sensors were evaluated by measuring their resistance using a WK6440B LCR meter. Sensors were affixed to a computer-controlled movement apparatus (TH16-57, Taida Machinery Equipment Co., Ltd.) using double-sided adhesive tape, allowing precise control over the application of mechanical strain and the rate of deformation.

To investigate the stability and repeatability of sensors, resistance measurements were taken over an extended period of time under different environmental conditions of temperature and humidity. Frequency-independent electrical characterization of the material was done with the help of an electrochemical workstation, providing comprehensive analyses for both static and dynamic loadings.

Gesture Recognition Experiment: For gesture recognition applications, Au-coated IL-infused nanowire-based sensors measuring 5.8 cm in length and 1.0 cm in width were securely attached to human fingers and wrists using medical-grade adhesive materials. Data acquisition was conducted for different gestures such as vertical finger movements, horizontal finger movements, and wrist bending motions.

During the experiments, specific angles of finger and wrist movements were maintained to standardize the tests. The cyclic sensitivity of the sensors was monitored, with each data collection session lasting ≈ 5 sec before transitioning to the next test. To evaluate the sensor responsiveness to different movement amplitudes, vertical motion angles were systematically varied and each angle was tested in five cyclic repetitions. Electrical signal changes were recorded to determine the correlation between movement angle and sensor output. Furthermore, the sensor capability to detect changes in gesture direction was assessed by alternating between vertical motion (six cycles) and horizontal motion (three cycles). Electrical data were captured throughout these transitions to evaluate the sensor's adaptability to complex and multidirectional movements.

All the experimental data was systematically stored and treated by validity checks to evaluate the reliability of the results. The wide range of datasets used provided a comprehensive performance test of the sensor in the recognition and response of hand gestures and movements, hence underlining its applications in wearable electronics, as well as human-machine interface technologies.

Ethics Approval and Human Research Participants: This research involving human participants was ethically cleared by Departments of Academic Ethics Committee at the Hebei University. All individuals received detailed participant information and provided written consent before taking part in the studies.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Author Contributions

S.P. and Y.Z. contributed equally to this work. All authors contributed to this work.

Data Availability Statement

The authors declare that the main data supporting the findings of this study are contained within the paper and the Supplementary Information files.

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